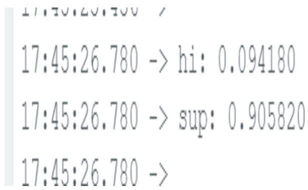
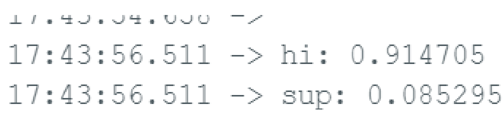
EE446
LAB 9
Report

1) Screenshots showing both ASL gestures correctly identified by the Arduino Serial Monitor

Sup identification


17:45:26.780 -> hi: 0.094180
17:45:26.780 -> sup: 0.905820
17:45:26.780 ->

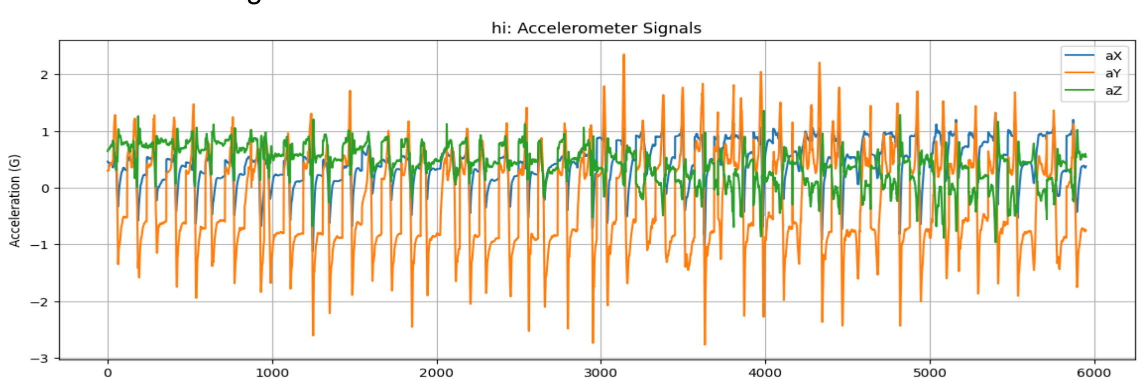
Hi identification


17:43:56.511 -> hi: 0.914705
17:43:56.511 -> sup: 0.085295

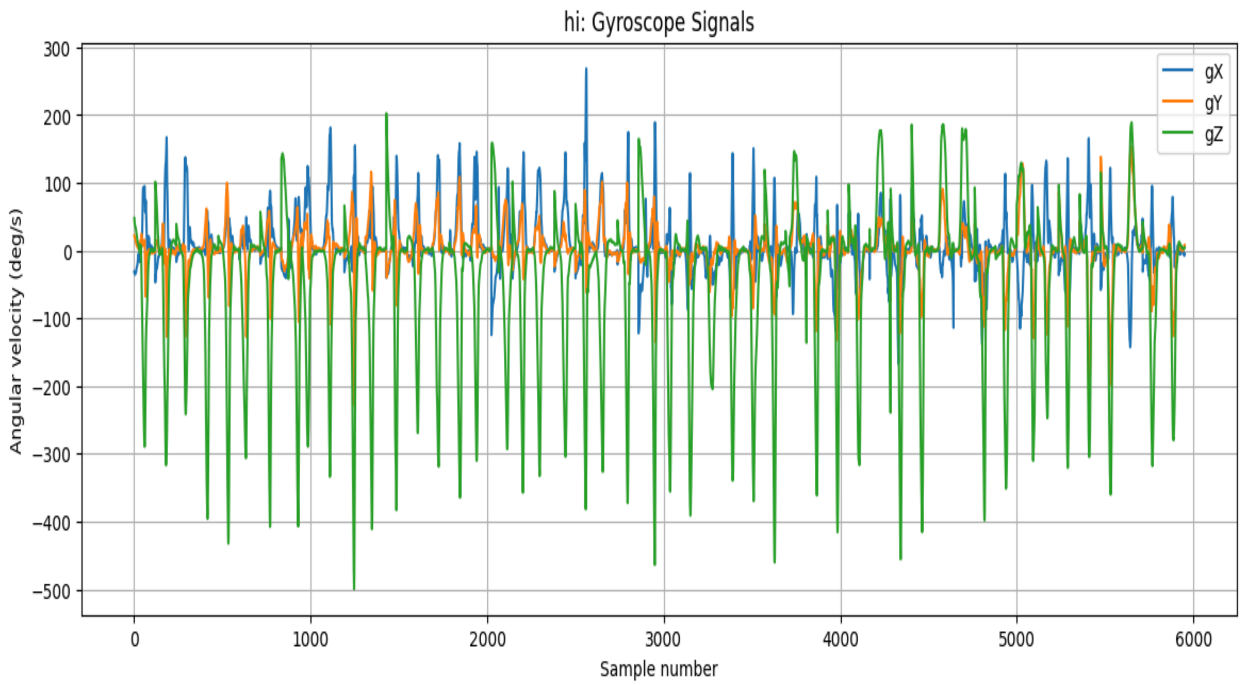
2) A brief explanation of the software issue fixed in the notebook

I changed the metric from MAE to Accuracy, because MAE is used for regression problems and doesn't give results for one-hot class labels. Since the output layer uses softmax over output classes and this is a classification task, accuracy directly tells me how often the model predicts the correct gesture. I also switched to Categorical Cross-Entropy because it properly measures how well the predicted probability distribution matches the true label and penalizes confident wrong predictions more effectively.

Plots from the notebook:
Hi accelerometer signals

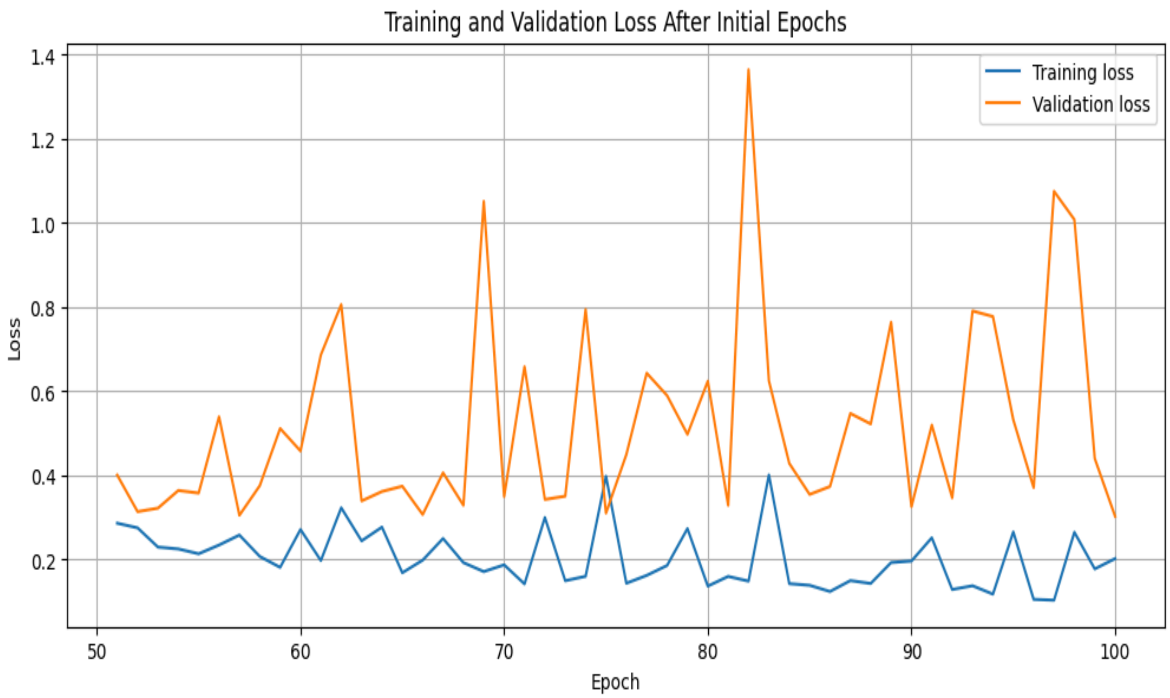


Hi gyroscope signals

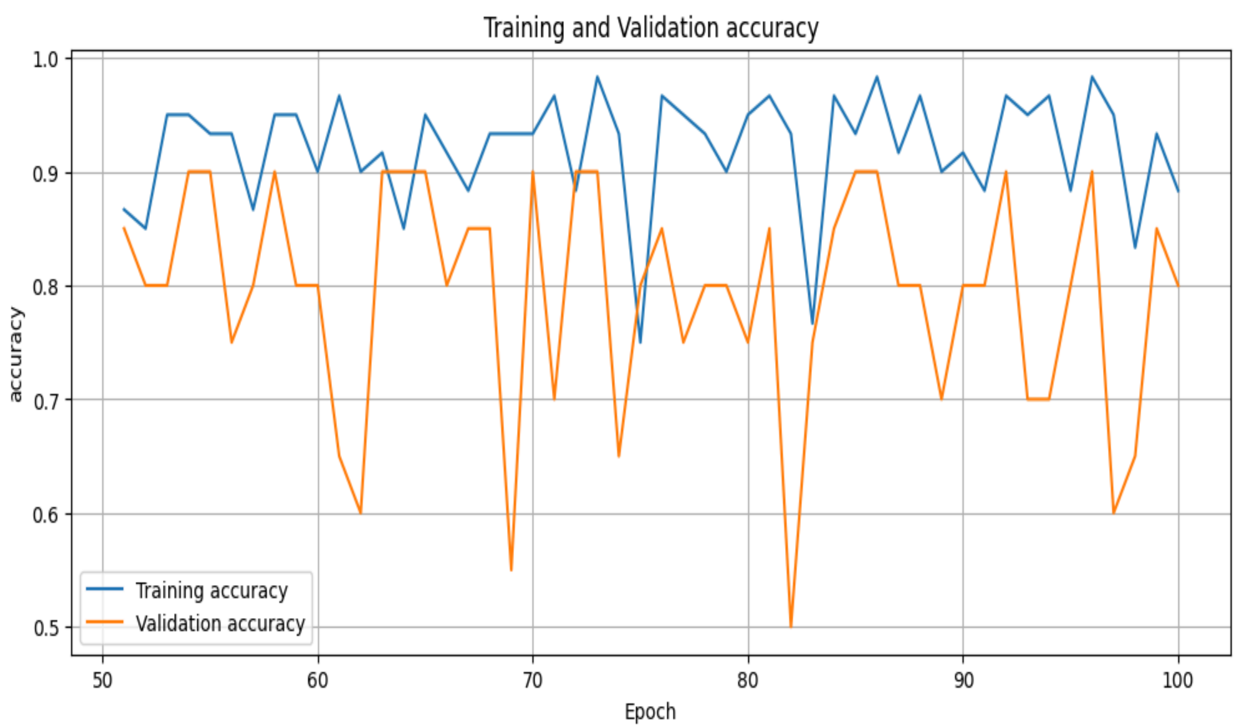


Train and test loss vs epoch plot

Train and test loss vs epoch plot for last 50 epochs



Train and test accuracy vs epoch plot



Predicted vs Actual gesture vs sample index